Lab setup

NetApp Solutions

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Lab setup

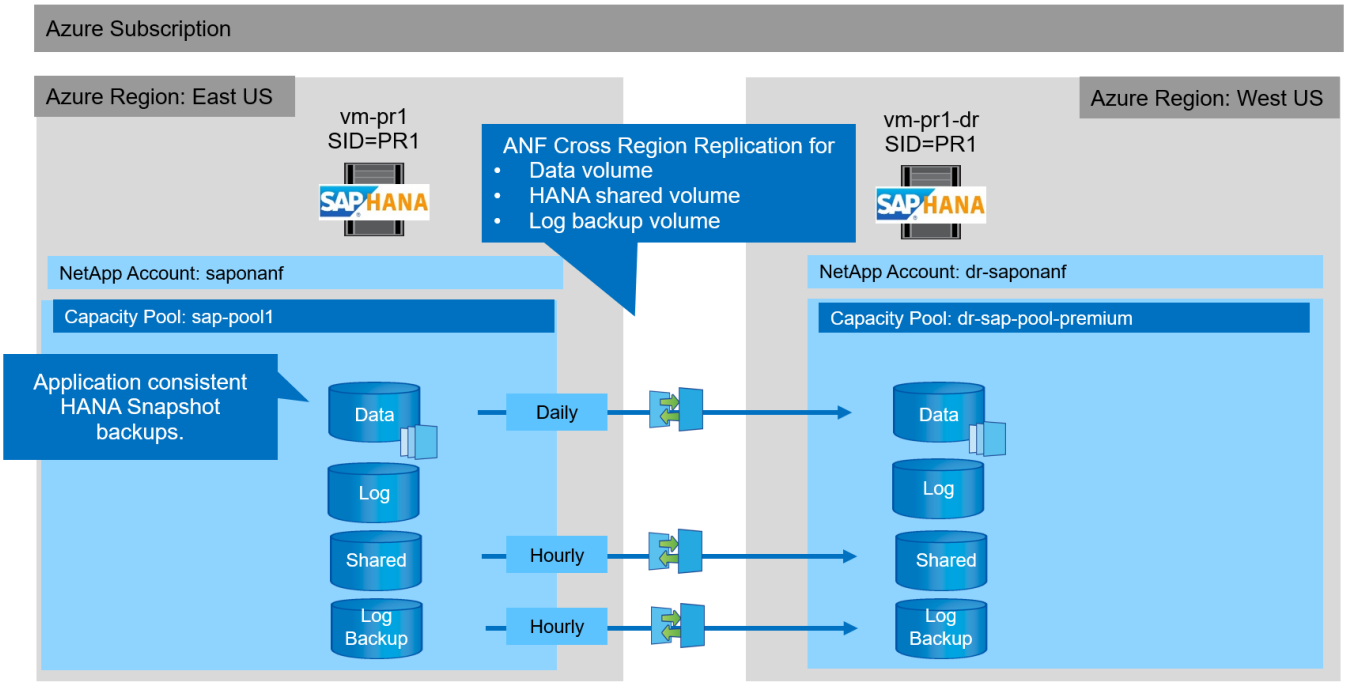
Solution validation has been performed with an SAP HANA single-host system. The Microsoft AzAcSnap Snapshot backup tool for ANF has been used to configure HANA application-consistent Snapshot backups. A daily data volume, hourly log backup, and shared volume replication were all configured. Disaster recover testing and failover was validated with a save point as well as with forward recovery operations.

The following software versions have been used in the lab setup:

- Single host SAP HANA 2.0 SPS5 system with a single tenant
- SUSE SLES for SAP 15 SP1
- AzAcSnap 5.0

A single capacity pool with manual QoS has been configured at the DR site.

The following figure depicts the lab setup.



Snapshot backup configuration with AzAcSnap

At the primary site, AzAcSnap was configured to create application-consistent Snapshot backups of the HANA system PR1. These Snapshot backups are available at the ANF data volume of the PR1 HANA system, and they are also registered in the SAP HANA backup catalog, as shown in the following two figures. Snapshot backups were scheduled for every 4 hours.

With the replication of the data volume using ANF Cross-Region Replication, these Snapshot backups are replicated to the disaster recovery site and can be used to recover the HANA database.

The following figure shows the Snapshot backups of the HANA data volume.

1-data-mnt00001)

PR1-data-mnt00001 (saponanf/sap-pool1/PR1-data-mnt00001) | Snapshots

Volume

Search (Ctrl+/)

Overview
Activity log
Access control (IAM)
Tags

Settings

Properties
Locks

Storage service

Mount instructions
Export policy
Snapshots
Replication

Monitoring

Metrics

Add snapshot Refresh

Search snapshots

Name	Location	Created
azacsnap__2021-02-12T145015-1799555Z	East US	02/12/2021, 03:49:48 PM
azacsnap__2021-02-12T145227-1245630Z	East US	02/12/2021, 03:51:24 PM
azacsnap__2021-02-12T145828-3863442Z	East US	02/12/2021, 03:58:01 PM
azacsnap__2021-02-16T134021-9431230Z	East US	02/16/2021, 02:39:18 PM
azacsnap__2021-02-16T134917-6284160Z	East US	02/16/2021, 02:48:55 PM
azacsnap__2021-02-16T135737-3778546Z	East US	02/16/2021, 02:56:32 PM
azacsnap__2021-02-16T160002-1354654Z	East US	02/16/2021, 04:59:40 PM
azacsnap__2021-02-16T200002-0790339Z	East US	02/16/2021, 08:59:42 PM
azacsnap__2021-02-17T000002-1753859Z	East US	02/17/2021, 12:59:32 AM
azacsnap__2021-02-17T040001-5454808Z	East US	02/17/2021, 04:59:31 AM
azacsnap__2021-02-17T080002-2933611Z	East US	02/17/2021, 08:59:40 AM

The following figure shows the SAP HANA backup catalog.

n-pr1 Instance: 01 Connected User: SYSTEM System Usage: Custom System - SAP HANA Studio

Help

SYSTEMDB@PR1 ... | Backup SYSTE... | SYSTEMDB@PR1 ... | SYSTEMDB@PR1 | SYSTEMDB@PR1 | Backup SYSTE... | SYSTEMDB@PR1 ... | SYSTEMDB@PR1 ... | SYSTEMDB@PR1 ...

Backup SYSTEMDB@PR1 (SYSTEM) PR1 SystemDB Last Update:9:07:38 AM

Overview | Configuration | Backup Catalog

Backup Catalog

Database: SYSTEMDB

Show Log Backups Show Delta Backups

Status	Started	Duration	Size	Backup Type	Destinatio...
	Feb 17, 2021 8:00:02 ...	00h 00m 42s	3.13 GB	Data Backup	Snapshot
	Feb 17, 2021 4:00:01 ...	00h 00m 35s	3.13 GB	Data Backup	Snapshot
	Feb 17, 2021 12:00:0...	00h 00m 36s	3.13 GB	Data Backup	Snapshot
	Feb 16, 2021 8:00:02 ...	00h 00m 34s	3.13 GB	Data Backup	Snapshot
	Feb 16, 2021 4:00:02 ...	00h 00m 38s	3.13 GB	Data Backup	Snapshot
	Feb 16, 2021 1:57:37 ...	00h 00m 32s	3.13 GB	Data Backup	Snapshot
	Feb 16, 2021 1:49:17 ...	00h 00m 32s	3.13 GB	Data Backup	Snapshot
	Feb 16, 2021 1:40:22 ...	00h 00m 34s	3.13 GB	Data Backup	Snapshot
	Feb 12, 2021 2:58:28 ...	00h 00m 32s	3.13 GB	Data Backup	Snapshot
	Feb 12, 2021 2:52:27 ...	00h 00m 32s	3.13 GB	Data Backup	Snapshot
	Feb 12, 2021 2:50:15 ...	00h 00m 32s	3.13 GB	Data Backup	Snapshot

Backup Details

ID:	1613141415533
Status:	Successful
Backup Type:	Data Backup
Destination Type:	Snapshot
Started:	Feb 12, 2021 2:50:15 PM (UTC)
Finished:	Feb 12, 2021 2:50:48 PM (UTC)
Duration:	00h 00m 32s
Size:	3.13 GB
Throughput:	n.a.
System ID:	
Comment:	Snapshot prefix: azacsnap Tools version: 5.0 Preview (20201214.65524)
Additional Information:	<ok>
Location:	/hana/data/PR1/mnt00001/

Host	Service	Size	Name	Source ...	EBID
vm-pr1	nameserver	3.13 GB	hdb00001	volume	azacsnap__2021-02-12T14501...

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